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1. What Is This Project About?

The core question

A company that sells subscription services (think Netflix, a phone plan, or a gym membership) wants to know:

> **"Which of our customers are likely to cancel their subscription in the near future?"**

When a customer cancels, it is called **churn**. Churn is expensive for the company because:

- It costs far more to find a new customer than to keep an existing one.
- Lost customers mean lost monthly revenue forever.
- If many customers leave at the same time, it signals a product or pricing problem.

What this project does

This project builds a complete analytics system that:

1. **Takes raw customer data** (age, spending, contract type, etc.)
2. **Trains a machine learning model** that predicts the probability that each customer will churn
3. **Exports the predictions and model metrics** in a format that Power BI can read
4. **Displays everything in a multi-page Power BI dashboard** so business managers can take action

Think of it as a **crystal ball for customer cancellations** — powered by data and statistics, not guesswork.

---

2. The Dataset — What Data Do We Have?

Source

The data comes from **Kaggle** (a popular data science competition website), published by Muhammad Shahid Azeem. It was split into two files:

File	Records
`customer_churn_dataset-training-master.csv`	440,833 rows
`customer_churn_dataset-testing-master.csv`	64,374 rows
Total combined	**505,207 rows**

Each row represents **one customer**. Each column tells us one fact about that customer.

The 12 Attributes (Columns) Explained

1. `CustomerID`

- **What it is:** A unique number that identifies each customer. Like a membership card number.
- **Type:** Number (float64)
- **Example values:** 2.0, 3.0, 4.0, ... up to 449,999
- **Why it matters:** Used only to link records together. It has **no predictive value** — knowing a customer's ID number cannot tell you whether they will churn. So it is **removed from the model** but kept in the output tables for lookup.
- **Note:** Values are floats (e.g. 2.0 instead of 2) because the dataset has a few missing rows and pandas uses float for integer columns with any null.

---

2. `Age`

- **What it is:** The customer's current age in years.

- **Type:** Number
- **Range:** 18 to 65
- **Average:** ~39 years
- **Why it matters:** Age can influence behaviour. Younger customers may switch services more frequently. Older customers with longer product histories may be more loyal or more set in their ways.
- **In the model:** This is the **second most important predictor** of churn (importance score: 458 out of 1,000+, ranking #2 overall).

---

3. `Gender`

- **What it is:** The customer's gender — either `Male` or `Female`.
- **Type:** Text (categorical)
- **Distribution:** ~57% Male, ~43% Female
- **Why it matters:** Some products or pricing structures may affect genders differently. The model learns if gender correlates with churn in this dataset.
- **In the model:** Ranked 8th in importance (score: 180). It has some influence but is far less important than financial and behavioral attributes.

---

4. `Tenure`

- **What it is:** How long the customer has been with the company, measured in **months**.
- **Type:** Number
- **Range:** 1 to 60 months (up to 5 years)
- **Average:** ~31 months
- **Why it matters:** This is a classic churn predictor. Customers who have just joined are most at risk because they have not yet built loyalty or habit. Long-tenure customers represent the stable, satisfied base.
- **In the model:** Ranked **4th** most important (score: 375). Newer customers churn far more.

---

5. `Usage_Frequency`

- **What it is:** How many times per month the customer uses the product or service. (Think number of logins, sessions, or visits.)
- **Type:** Number
- **Range:** 1 to 30 (times per month)
- **Average:** ~16
- **Why it matters:** Low engagement is a strong signal. A customer who barely uses a service they are paying for is much more likely to decide it is not worth the money and cancel.
- **In the model:** Ranked **5th** most important (score: 304).

---

6. `Support_Calls`

- **What it is:** How many times the customer called customer support in the past period.
- **Type:** Number
- **Range:** 0 to 10 calls
- **Average:** ~3.6 calls
- **Why it matters:** Customers who contact support multiple times are often frustrated. High support call counts suggest unresolved problems, which drives dissatisfaction and eventual cancellation.
- **In the model:** Ranked **6th** most important (score: 292). High support calls = higher churn risk.

---

7. `Payment_Delay`

- **What it is:** The number of days (on average) that the customer is **late paying their bill**.
- **Type:** Number
- **Range:** 0 to 30 days late
- **Average:** ~13 days
- **Why it matters:** Late payments indicate financial stress or declining commitment to the service. A customer who consistently pays late is signaling they may soon stop paying altogether.
- **In the model:** Ranked **3rd** most important (score: 407). One of the strongest churn signals.

---

8. `Subscription_Type`

- **What it is:** Which tier of service the customer is on.
- **Type:** Text (categorical)
- **Values:** `Basic`, `Standard`, `Premium`
- **Distribution (approx):** Standard ~34%, Premium ~34%, Basic ~32%
- **Why it matters:** Customers on lower-tier plans have paid less, have less invested in the service, and may be more likely to leave. Premium customers who get more value may be more loyal.
- **In the model:** The three tiers are converted to separate indicator columns. Combined importance: relatively low (ranked 9–13th area).

---

9. `Contract_Length`

- **What it is:** The billing cycle the customer is on — how often they are charged and committed.
- **Type:** Text (categorical)
- **Values:** `Monthly`, `Quarterly`, `Annual`
- **Distribution:** Annual ~40%, Quarterly ~40%, Monthly ~20%

- **Why it matters:** Month-to-month customers can leave at any time with no penalty. Annual subscribers have already committed for a year, so their immediate churn risk is much lower. This is one of the most important business levers for reducing churn.
- **In the model:** Converted to indicator columns. Monthly contract customers show significantly higher churn.

---

10. `Total_Spend`

- **What it is:** The total amount of money the customer has spent with the company (in dollars/currency units).
- **Type:** Number
- **Range:** $100 to $1,000
- **Average:** ~$632
- **Why it matters:** Spend reflects both loyalty (long-term customers spend more) and value. High-spend customers are priority retention targets because losing them hurts revenue most.
- **In the model:** The **single most important predictor** of churn (importance score: 467, ranking #1). High spenders who churn represent the biggest business risk.

---

11. `Last_Interaction`

- **What it is:** How many days ago the customer last interacted with the company (logged in, made a purchase, contacted support, etc.).
- **Type:** Number
- **Range:** 1 to 30 days ago
- **Average:** ~14 days
- **Why it matters:** A customer who has not touched the product for weeks is "going cold." Recent interaction is a strong signal of ongoing engagement and lower churn risk.
- **In the model:** Ranked **6th (tied with Support_Calls)** (score: 292).

---

12. `Churn` — The Target Column

- **What it is:** Whether the customer actually churned (left the company).
- **Type:** Number representing Yes/No
- **Values:** `1.0` = churned (left), `0.0` = retained (stayed)
- **Distribution:** ~57% of customers churned, ~43% were retained
- **Why it matters:** This is the **label we are trying to predict**. The machine learning model learns patterns from all the other columns to predict this one.
- **Important note:** The fact that 57% of customers churned means this dataset has slightly more churners than non-churners. This is a moderately imbalanced dataset (many real-world churn datasets are much more imbalanced).

Visual Summary of the 12 Columns

```
■■■
```

Why combine them? The "train/test" split in the file names refers to the original Kaggle competition setup. For our project, we want the full customer population in the database, so we merge them and let PyCaret create its own internal train/test split.

Step 2: Column Name Standardization

Raw column names have spaces (e.g. `"Usage Frequency"`). The `clean_column_names()` function converts all names to **snake_case** (underscores instead of spaces):

Raw Name	Cleaned Name
`Usage Frequency`	`Usage_Frequency`
`Support Calls`	`Support_Calls`
`Payment Delay`	`Payment_Delay`
`Subscription Type`	`Subscription_Type`
`Contract Length`	`Contract_Length`
`Last Interaction`	`Last_Interaction`

Why? Power BI, Python, and SQL all work better with column names that have no spaces. Consistent naming prevents errors throughout the pipeline.

Step 3: String Cleanup

All text columns are stripped of leading/trailing spaces. Common null-like text values (`"NA"`, `"N/A"`, `""`, `"None"`) are replaced with proper `NaN` (missing value marker).

Why? Raw data often has invisible formatting issues. A value like `" Male "` (with spaces) would be treated as different from `"Male"`. Cleaning standardizes this.

Step 4: Target Column Detection

The code automatically looks for a column named `Churn` and marks it as the target (what we want to predict). This is configured in `src/config.py` as a list of candidate names.

The Cleaned Table Design (Parquet Schema)

After cleaning, the data is saved to three formats in `data/processed/parquet/`:

`fact_churn_base.parquet` — The full cleaned dataset

All 505,207 rows with all 12 original columns (cleaned). Used as the single source of truth.

```
Columns (12):
CustomerID float64 Unique customer identifier
Age float64 Customer age
Gender object Male / Female
Tenure float64 Months as a customer
Usage_Frequency float64 Monthly usage count
Support_Calls float64 Support calls made
Payment_Delay float64 Days late on payment
Subscription_Type object Basic / Standard / Premium
Contract_Length object Monthly / Quarterly / Annual
Total_Spend float64 Total lifetime spend
Last_Interaction float64 Days since last interaction
Churn float64 1.0 = churned, 0.0 = retained
```

`fact_churn.parquet` — The Power BI fact table

Same as above but further shaped for Power BI analysis. This is the central table in the data model.

`dim_churn.parquet` — Dimension table for churn labels

```
Columns:
churn_label object Distinct churn values (0.0 or 1.0)
churn_key int64 Surrogate key (1, 2)
```

Why a separate table? This is **star schema design** — a best practice for Power BI performance. Dimension tables hold unique lookup values; fact tables hold the raw measurements.

`model_features.parquet` — Features used in training

All columns except `CustomerID` and `Churn`. Used internally by the model training pipeline.

`model_target.parquet` — The target variable only

Just the `Churn` column. Used internally during training.

---

4. The Problem We Are Solving and How

The Business Problem

A subscription service is losing customers. Management wants to know:

1. **Who is about to leave?** (Risk identification)

2. **How confident are we?** (Model reliability)

3. **Where to focus retention budget?** (Prioritization)

4. **If we save X% of at-risk customers, how much revenue do we recover?** (Scenario planning)

Why Not Just Use Simple Rules?

You might think: "Just flag everyone who has not logged in for 2 weeks." The problem is that single rules miss most cases and generate too many false alarms. In reality, churn is caused by a **combination** of factors — a customer who has low usage AND is on a monthly contract AND has high payment delay is far more likely to churn than someone with only one of those signals.

Machine learning automatically finds these **multi-factor combinations** from hundreds of thousands of historical examples.

The Solution Approach

| Stage | What Happens |

|---|---|

| 1. Data | Historical records: we know who churned and who stayed |

| 2. Model | Machine learns patterns that distinguish churners from non-churners |

| 3. Score | Model scores every current customer: "X% chance of churning" |

| 4. Segment | Customers are bucketed into Low / Medium / High risk |

| 5. Act | Business targets high-risk customers with retention campaigns |

What "Probability" Means Here

The model does not just say "will churn" or "will not churn." It gives a **probability score between 0 and 1**:

```

Score 0.05 → 5% chance of churn → Low Risk

Score 0.55 → 55% chance of churn → Medium Risk

Score 0.85 → 85% chance of churn → High Risk
```

The business decides a **threshold** (default: 0.50). Everyone above the threshold is predicted to churn. We explored thresholds from 0.05 to 0.95 to find the optimal balance.

---

5. Feature Engineering — Preparing Data for the Model

What Is Feature Engineering?

Machine learning models are picky. They need:

- Numbers (not text like "Male" or "Annual")
- No identifier columns (CustomerID adds noise, not signal)
- Optional extra features that might help the model

The `feature_engineering()` function in `src/data/pipeline.py` handles this.

Step 1: Remove the CustomerID

`CustomerID` is dropped from the model input before training.

Why? Customer IDs are just arbitrary numbers. The model might notice that customer #502 churned and customer #503 did not — but that pattern is noise, not signal. IDs leak no real information about churn behavior. Including them would cause **overfitting** (the model memorizing IDs instead of learning patterns).

Step 2: Two Helper Columns Are Added

```python
numeric_feature_count = how many of the numeric columns are non-null for this row

categorical_feature_count = how many of the text columns are non-null for this row
```

Why? These count how "complete" a customer's record is. A customer with many missing fields is harder to predict. These summary counts give the model a way to factor in data completeness. In practice, the final importance scores show these two features ranked last (importance: 0), meaning the data was too complete for them to matter — which is actually a sign of data quality.

Step 3: Encoding Categorical Columns

After `feature_engineering()`, PyCaret handles encoding automatically during setup. Text values like `"Monthly"` and `"Annual"` are converted to numeric indicator (dummy) columns:

```
Contract_Length_Monthly 1 if Monthly, 0 otherwise

Contract_Length_Quarterly 1 if Quarterly, 0 otherwise

Contract_Length_Annual 1 if Annual, 0 otherwise

Subscription_Type_Basic 1 if Basic, 0 otherwise

Subscription_Type_Standard 1 if Standard, 0 otherwise

Subscription_Type_Premium 1 if Premium, 0 otherwise
```

Why not just use 1/2/3? If you encode "Monthly=1, Quarterly=2, Annual=3", the model would think Annual (3) is "three times more" than Monthly (1) — which is nonsense. Separate indicator columns avoid this false ordering.

Final Feature List Sent to the Model (16 Features)

#	Feature	Origin
1	Age	Original
2	Tenure	Original
3	Usage_Frequency	Original
4	Support_Calls	Original
5	Payment_Delay	Original
6	Total_Spend	Original
7	Last_Interaction	Original
8	Gender (encoded)	Original (encoded)
9	Contract_Length_Monthly	Engineered from Contract_Length
10	Contract_Length_Quarterly	Engineered from Contract_Length
11	Contract_Length_Annual	Engineered from Contract_Length
12	Subscription_Type_Basic	Engineered from Subscription_Type
13	Subscription_Type_Standard	Engineered from Subscription_Type
14	Subscription_Type_Premium	Engineered from Subscription_Type
15	numeric_feature_count	Engineered helper
16	categorical_feature_count	Engineered helper

---

6. How We Used PyCaret — Step by Step

What Is PyCaret?

PyCaret is a Python library that makes machine learning accessible. Instead of writing hundreds of lines of scikit-learn code to train, evaluate, and compare models, PyCaret lets you do it in a handful of lines.

Think of PyCaret as a **recipe book** — it handles all the standard steps (scaling, encoding, cross-validation, metric calculation) and lets you compare dozens of algorithms at once.

Step 1: Setup (`setup()`)

```python
exp = setup(
data=model_df, # the cleaned dataset
target='Churn', # what we want to predict
session_id=42, # random seed (for reproducibility)
train_size=0.8, # 80% of data for training, 20% for testing
fold=5, # 5-fold cross-validation
normalize=True, # scale numeric values to 0–1 range
remove_multicollinearity=True, # drop features that are near-duplicates of each other
multicollinearity_threshold=0.9,
verbose=False
)
```

What each setting means:

Setting	Value	Why
`train_size=0.8`	80/20 split	Standard practice. Reserve 20% to test on data the model never saw.
`fold=5`	5-fold CV	Train/evaluate 5 times on different slices. More reliable than a single split.
`normalize=True`	Yes	Ensures 30-day `Tenure` and $1000 `Total_Spend` are on the same scale.
`remove_multicollinearity=True`	Yes	Drops redundant features. Improves stability.
`session_id=42`	42	Locks random number generator so results are reproducible.

Why we sampled 60,000 rows for training: The full dataset has 505,207 rows. Training on all of them would take a very long time. We sampled 60,000 rows in a **stratified** way (maintaining the same 57%/43% churn ratio) to get fast, representative results.

Step 2: Compare Models (`compare_models()`)

```python
best_models = compare_models(n_select=5, sort='AUC')
```

PyCaret trains and evaluates **15+ different algorithms** automatically. We asked for the top 5 sorted by AUC (AUC explained below).

The algorithms tested include:

- Logistic Regression (simple linear model)
- Random Forest (ensemble of decision trees)

- Gradient Boosting (sequential tree improvement)
- LightGBM (fast, optimized gradient boosting)
- Neural Network, KNN, SVM, and more

Why AUC for sorting? AUC (Area Under the ROC Curve) measures how well the model ranks churners above non-churners across all possible thresholds. A score of 1.0 is perfect; 0.5 is no better than random guessing.

Step 3: The Winning Model — LightGBM

LightGBM (Light Gradient Boosting Machine) ranked first with these cross-validated scores:

Metric	Score	What It Means
AUC	**0.9541**	Excellent discrimination. Ranks 95.4% of churners above non-churners.
Recall	**0.9956**	Catches 99.56% of all actual churners. Almost none missed.
F1	**0.9435**	Balanced score between precision and recall. Strong overall.
Precision	**0.8965**	When it says "will churn", 89.65% of those predictions are correct.
Accuracy	**0.9338**	93.38% of all predictions are correct.

Why LightGBM wins: It uses a technique called **gradient boosting** — it builds many small decision trees sequentially, each one fixing the errors of the previous. The result is a model that captures complex, non-linear interactions (e.g. the combination of "6+ support calls AND payment delay > 20 days AND monthly contract" being a very high risk signal).

Step 4: Understanding the Metrics

Precision vs. Recall — Why Both Matter

Imagine a doctor screening for a disease. There are two mistakes they can make:

Mistake	In Our Context	Cost
False Negative (miss a sick person)	Predict "not churning" for someone who actually churned	Customer leaves without us trying to save them
False Positive (alarm for a healthy person)	Predict "churning" for someone who stays	Wasted retention budget on someone happy

The balance between these two error types is controlled by the **threshold**:

- Lower threshold (e.g. 0.15) = catch more churners but also alarm on more non-churners
- Higher threshold (e.g. 0.70) = only flag very likely churners, but miss some

Our model uses **threshold=0.15** as optimal:

```
At threshold 0.15:
Recall = 0.9996 (catches almost everyone about to churn)
Precision = 0.8962 (89.6% of "will churn" alerts are genuine)
F1 = 0.9451 (best balance between the two)
```

Why 0.15 and not the default 0.50? In churn prevention, missing a churner (false negative) is more costly than incorrectly targeting a happy customer (false positive). A low threshold intentionally catches more churners, accepting slightly more false alarms, because intervening with a staying customer wastes a small discount — but missing a churning customer loses all future revenue.

Step 5: Feature Importance Export

After the best model is identified, we extract **which features it found most useful**:

Rank	Feature	Importance Score
1	Total_Spend	467
2	Age	458
3	Payment_Delay	407
4	Tenure	375
5	Usage_Frequency	304
6	Support_Calls	292
7	Last_Interaction	292
8	Gender	180
9	Contract_Length_Monthly	105
...	(Subscription type splits)	18–28

Interpretation: Total spend, age, and payment delay are the three most powerful predictors. Monthly contract is the most important subscription signal. Exactly which subscription tier matters less.

This is saved to `dim_feature_importance.parquet` and displayed as a bar chart in the dashboard.

Step 6: Full Dataset Scoring

After model training, we run the model against **all 505,207 customers** (not just the test set) to generate a churn score for every customer. This becomes the `fact_customer_scores` table.

Each row in that table contains:

- The customer's churn probability (0.0 to 1.0)
- Their predicted churn status (0 or 1, using threshold 0.50 default)
- Their risk segment: Low Risk / Medium Risk / High Risk

---

7. All Calculated Columns and Measures in Power BI

Power BI has two types of calculations:

- **Calculated Columns:** Stored permanently in the table, computed row by row when data is refreshed.
- **Measures (DAX):** Calculated dynamically when a visual is rendered, respecting any filters applied.

All our calculations are DAX Measures stored in the `_Measures` table. There are no custom calculated columns. Here is every measure, explained:

Group 1: Core Volume Metrics

`Total Customers`

```dax
Total Customers = DISTINCTCOUNT('fact_churn'[CustomerID])
```

What: Counts unique customer IDs.

Why: The most fundamental KPI. Everything else is expressed as a percentage or proportion of this number.

`Total Churned`

```dax
Total Churned =
CALCULATE(
DISTINCTCOUNT('fact_churn'[CustomerID]),
KEEPFILTERS('fact_churn'[Churn] = 1)
)
```

What: Counts only customers who churned (Churn = 1.0).

Why: The raw count of lost customers. Combined with Total Customers, this gives you Churn Rate.

KEEPFILTERS: Ensures that any dashboard filter (e.g. "Annual contracts only") is respected.

`Total Retained`

```dax
Total Retained = [Total Customers] - [Total Churned]
```

What: Customers who stayed.

Why: Paired with Total Churned for donut chart splits and context.

`Churn Rate %`

```dax
Churn Rate % = DIVIDE([Total Churned], [Total Customers], 0)
```

What: The percentage of customers who churned.

Why: The headline executive KPI. Formatted as a percentage (e.g. 56.7%). `DIVIDE` is used instead of `/` to safely handle division by zero.

`Retention Rate %`

```dax
Retention Rate % = 1 - [Churn Rate %]
```

What: The percentage of customers who stayed.

Why: Same information as churn rate but framed positively. Useful for leadership who prefer to see growth metrics.

Group 2: Financial Metrics

`Total Spend`

```dax
Total Spend = SUM('fact_churn'[Total_Spend])
```

What: Sum of all customer spending.

Why: The total revenue base. Establishes the scale of the business.

`Spend at Risk`

```dax
```

```
Spend at Risk = CALCULATE(
[Total Spend],
KEEPFILTERS('fact_churn'[Churn] = 1)
)
```

What: Total spend by churned customers only.

Why: Shows the actual revenue already lost. This is the dollar value of the problem.

`Average Spend Per Customer`

```dax
Average Spend Per Customer = DIVIDE([Total Spend], [Total Customers], 0)
```

What: Average revenue per customer.

Why: Normalizes spend for comparison across segments. A high-churn segment with low average spend is less urgent than a moderate-churn segment with very high average spend.

`Churn Rate by Contract %`

```dax
Churn Rate by Contract % = DIVIDE([Total Churned], [Total Customers], 0)
```

What: Same calculation as Churn Rate % — this measure is applied in the context of contract type filters.

Why: Used specifically in charts that break down churn by contract length, so it adopts whatever filter context the visual applies.

Group 3: Risk and Scoring Metrics

These measures come from the **`fact_customer_scores`** table, which contains the model's predictions for every customer.

`High Risk Customers`

```dax
High Risk Customers = CALCULATE(
DISTINCTCOUNT('fact_customer_scores'[CustomerID]),
KEEPFILTERS('fact_customer_scores'[risk_segment] = "High Risk")
)
```

```

**What:** Count of customers the model scored above 0.70 churn probability.

**Why:** The model segments customers into Low (0–40%), Medium (40–70%), and High (70–100%) risk. This KPI counts the most urgent group — the ones most likely to leave soon.

---

#### `High Risk Revenue`

```dax
High Risk Revenue = CALCULATE(
SUM('fact_customer_scores'[Total_Spend]),
KEEPFILTERS('fact_customer_scores'[risk_segment] = "High Risk")
)
```

What: Total spend from high-risk customers.

Why: Not all churners are equal. A high-risk customer who spends $1,000/year matters much more than one who spends $100. This metric connects risk to revenue impact.

Group 4: Scenario / "What-If" Metrics

These measures help the business plan retention campaigns:

`Potential Saves (10%)`

```dax
Potential Saves (10%) = ROUND([High Risk Customers] * 0.10, 0)
```

What: If we intervene with all high-risk customers and convert 10% of them, how many do we save?

Why: Conservative scenario. Retention campaigns typically have 5–20% success rates.

`Potential Saves (20%)`

```dax
Potential Saves (20%) = ROUND([High Risk Customers] * 0.20, 0)
```

What: Optimistic scenario — 20% conversion rate.

Why: Paired with the 10% scenario so decision-makers can see a range.

`Potential Spend Saved (20%)`
```

```dax
Potential Spend Saved (20%) = [High Risk Revenue] * 0.20
```

What: If we retain 20% of high-risk customers, how much revenue do we recover?

Why: Converts the customer count scenario into a dollar-value recovery estimate. Helps justify the investment in retention campaigns.

`Potential Spend Saved (30%)`

```dax
Potential Spend Saved (30%) = [High Risk Revenue] * 0.30
```

What: Most optimistic scenario — 30% conversion.

Why: Upper bound for budget planning.

Group 5: Model Performance Metrics (Explainability KPIs)

These come from `dim_model_metrics`, a one-row table storing the best model's performance numbers.

`Best AUC`

```dax
Best AUC = MAX('dim_model_metrics'[auc])
```

What: The model's AUC score — **0.9541**.

Why: The headline model quality metric. Tells stakeholders how reliable the model is. Shows on the Explainability page. MAX is used because there is one row, but the function safely handles single-row tables.

`Best Recall`

```dax
Best Recall = MAX('dim_model_metrics'[recall])
```

What: **0.9956** — the model catches 99.56% of all actual churners.

Why: For churn prevention, missing a churner is the worst error. High recall = few missed churners.

`Best F1`

```dax
Best F1 = MAX('dim_model_metrics'[f1])
```

What: **0.9435** — the harmonic mean of precision and recall.

Why: F1 is the single-number summary of model balance. It cannot be gamed by just predicting everyone churns (which would maximise recall but destroy precision).

`Best Precision`

```dax
Best Precision = MAX('dim_model_metrics'[precision])
```

What: **0.8965** — when the model says a customer will churn, it is right 89.65% of the time.

Why: High precision means retention budget is mostly spent on genuinely at-risk customers.

`Best Accuracy`

```dax
Best Accuracy = MAX('dim_model_metrics'[accuracy])
```

What: **0.9338** — 93.38% of all predictions (churn and non-churn) are correct.

Why: Overall correctness. Less informative than AUC/F1 for imbalanced problems, but intuitive for non-technical stakeholders.

`Best Threshold`

```dax
Best Threshold = MAX('dim_model_metrics'[best_threshold])
```

What: **0.15** — the probability cutoff that gives the best F1 score.

Why: Shows stakeholders the operational setting that was used to generate predictions. Explains why the model seems "aggressive" in flagging customers.

`Best Threshold Recall`, `Best Threshold Precision`, `Best Threshold F1`

```dax
Best Threshold Recall = MAX('dim_model_metrics'[best_threshold_recall])
Best Threshold Precision = MAX('dim_model_metrics'[best_threshold_precision])
```

Best Threshold F1 = MAX('dim_model_metrics'[best_threshold_f1])
```

**What:** The recall (0.9996), precision (0.8962), and F1 (0.9451) specifically at the optimal threshold of 0.15.

**Why:** The standard metrics above use threshold 0.50. These show the performance at the threshold actually used in scoring, giving a more accurate picture of operational performance.

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## 8. Every Dashboard Page — Charts, KPIs, and Why They Exist

The dashboard has **4 pages**, each answering a different business question.

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### Page 1: Executive Overview

**Audience:** CEO, VP, senior management

**Question it answers:** "How bad is our churn problem in numbers?"

#### KPI Cards (top row)

| KPI | Measure | Why |
|---|---|---|
| **Total Customers** | `Total Customers` | Establishes the scale of the customer base |
| **Churn Rate %** | `Churn Rate %` | The headline problem metric — is it getting better or worse? |
| **Revenue at Risk** | `Spend at Risk` | Puts a dollar figure on the churn problem |
| **Retention Rate %** | `Retention Rate %` | Positive framing for leadership |

#### Charts

| Chart | Type | Data | Why |
|---|---|---|---|
| **Monthly Churn Trend** | Line / Area | fact_churn over time | Shows if churn is improving, worsening, or seasonal |
| **Churn by Contract** | Clustered Bar | fact_churn grouped by Contract_Length | Reveals if monthly contracts are much worse than annual |
| **Retained vs Churned** | Donut | Total Retained vs Total Churned | Quick visual split of the overall customer base |
| **Churn by Region** | Map or Bar | fact_churn by geographic segment | Identifies if any location has an outsized problem |
| **Top Drivers** | Horizontal Bar | dim_feature_importance | Shows executives which factors drive churn most — Total Spend and Age lead |

---
```

Page 2: Risk Segments

Audience: Data analysts, CRM managers

Question it answers: "Which customer groups are most at risk, and how concentrated is the problem?"

KPI Cards (top row)

| KPI | Measure | Why |

|---|---|---|

| **High Risk Customers** | `High Risk Customers` | How many are in the danger zone right now? |

| **High Risk Revenue** | `High Risk Revenue` | What revenue is at stake from high-risk customers? |

| **Avg Churn Probability** | Average of fact_customer_scores[churn_probability] | Overall model confidence across the population |

| **Priority Segment** | Contextual | Which segment needs the most urgent attention? |

Charts

| Chart | Type | Data | Why |

|---|---|---|---|

| **Heatmap: Tenure × Contract** | Matrix heatmap | fact_customer_scores | Shows the intersection of two key risk factors — new monthly customers are highest risk |

| **Tree Map: Churn by Segment** | Tree Map | fact_customer_scores | Each rectangle = a segment. Bigger = more customers. Color = churn rate. Instantly shows concentration |

| **Segment Drill Table** | Detailed Table | fact_customer_scores | Row-level detail for analysts to filter and explore specific risk profiles |

| **Churn Rate by Gender** | Horizontal Bar | fact_churn | Is gender a significant differentiator? |

| **Spend at Risk by Contract Length** | Column Chart | fact_churn | Monthly vs Quarterly vs Annual — where is the financial exposure concentrated? |

Page 3: Retention Opportunities

Audience: Marketing team, retention specialists

Question it answers: "If we act now, how many customers can we save and how much money is that worth?"

KPI Cards (top row)

| KPI | Measure | Why |

|---|---|---|

| **Potential Saves (10%)** | `Potential Saves (10%)` | Conservative: how many customers saved at 10% conversion |

| **Potential Saves (20%)** | `Potential Saves (20%)` | Balanced: 20% conversion scenario |

| **Potential Spend Saved (20%)** | `Potential Spend Saved (20%)` | Revenue recovered in the balanced scenario |

| **Potential Spend Saved (30%)** | `Potential Spend Saved (30%)` | Aggressive: 30% conversion scenario |

Charts

| Chart | Type | Data | Why |

|---|---|---|---|

| **Saves % by Tenure Group** | Grouped Bar | fact_customer_scores + scenarios | Shows which tenure cohort (new vs mid vs long-term) has the most potential customers to save |

| **Spend Saved by Contract** | Column Chart | Scenario metrics by contract type | Prioritizes retention campaigns: annual contract customers save more revenue per save |

| **Retention Action Table** | Full-width Table | fact_customer_scores | Segment-level action priorities — which combinations of risk + value to target first |

Page 4: Model Explainability

Audience: Data scientists, technically curious managers, compliance/audit

Question it answers: "Is the model good? How does it work? Can we trust the predictions?"

This page exists because **AI models that cannot be explained cannot be trusted**. Before a business acts on a model's predictions, stakeholders need to verify that the model is genuinely learning patterns and not making lucky guesses.

KPI Cards (top row)

| KPI | Measure | Value | Why |

|---|---|---|---|

| **Best AUC** | `Best AUC` | 0.9541 | Overall discriminative power. 95%+ is excellent. |

| **Best Recall** | `Best Recall` | 0.9956 | We catch almost all churners. Low miss rate. |

| **Best F1** | `Best F1` | 0.9435 | Balanced performance. Strong combined score. |

| **Inference Threshold** | `Best Threshold` | 0.15 | The actual cutoff being used for predictions. |

Charts (3×2 Grid)

Row 1:

| Chart | Type | X-axis | Y-axis | Why |

|---|---|---|---|---|

| **Score Distribution by Risk Segment** | Clustered Column | risk_segment (Low/Med/High) | Count of CustomerID | Shows how many customers fall in each risk bucket. A healthy model produces a spread, not all in one bucket. |

| **Threshold Trade-off (F1 / Recall)** | Dual-line | threshold (0.05 to 0.95) | Average F1 and Average Recall | Shows how performance changes as you raise or lower the prediction cutoff. The crossover point guides threshold selection. |

| **Confusion Matrix** | Matrix Visual | predicted_label | actual_label | The fundamental 2×2 table showing True Positives, False Positives, True Negatives, False Negatives. |

Row 2:

| Chart | Type | X-axis | Y-axis | Why |

|---|---|---|---|---|

| **ROC Curve** | Line / Scatter | fpr (False Positive Rate) | tpr (True Positive Rate) | The classic model quality curve. A perfect model hugs the top-left. AUC 0.95 = the curve is very close to perfect. |

| **Precision-Recall Curve** | Line / Scatter | recall | precision | More informative than ROC for imbalanced datasets. Shows the precision/recall tradeoff across all thresholds. |

| **Threshold Metrics Table** | Table | threshold | precision, recall, f1, accuracy, predicted_churners | The full scorecard at each threshold. Allows manual inspection of the setting that was selected. |

Understanding the Confusion Matrix

The Confusion Matrix is the most important explainability visual. It shows four types of predictions:

```

PREDICTED: Retained PREDICTED: Churned

ACTUAL: Retained [True Negative TN] [False Positive FP]

ACTUAL: Churned [False Negative FN] [True Positive TP]
```

| Cell | Name | Meaning |

|---|---|---|

| **TN** | True Negative | Customer stayed — model correctly said "stay." Good. |

| **FP** | False Positive | Customer stayed — model wrongly said "churn." Wasted retention effort. |

| **FN** | False Negative | Customer churned — model wrongly said "stay." Missed opportunity. Most costly error. |

| **TP** | True Positive | Customer churned — model correctly said "churn." Intervention was justified. |

Our model has very few False Negatives (high recall), meaning it rarely misses actual churners — the goal was achieved.

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9. Key Insights Found and How They Can Help

Insight 1: Total Spend is the Biggest Churn Predictor

Finding: The single strongest predictor of churn is not what you might expect — it is Total Spend (importance: 467), ahead of age, payment behaviour, and contract type.

Interpretation: Customers who have spent more are actually at higher churn risk in this dataset. This seems counterintuitive, but it makes sense: high spenders who are dissatisfied feel the "pain" of their bill more acutely and are more motivated to search for alternatives. They are high-value customers who made a cost-conscious decision.

Business action: Do not assume high-spend customers are loyal. Proactively reach out to high-spend customers showing any other warning signs. Offer them value justification — features they are paying for but not using.

Insight 2: Payment Delay is a Very Strong Early Warning Signal

Finding: Payment Delay is the 3rd most important predictor (importance: 407). Customers with even moderate delays (15+ days) are disproportionately represented in the churned group.

Interpretation: A late payment is a behavioural signal — the customer is either in financial difficulty or is disengaging. Both lead to cancellation. The payment delay is visible days or weeks before the actual churn event.

Business action: Create an automated trigger: when any customer exceeds 15 days of payment delay, immediately flag them for outreach. Offer payment flexibility, not penalties. This is an entirely preventable churn category.

Insight 3: Monthly Contract Customers Are the Most Vulnerable

Finding: In the feature importance ranking, `Contract_Length_Monthly` scores 105 — far ahead of Quarterly (26) and Annual (23). Monthly customers have the highest churn concentration.

Interpretation: Monthly contracts offer zero switching cost. Customers can (and do) cancel at end of each billing cycle with no penalty. Annual customers have already committed financially and are behaviorally more stable.

Business action:

1. Offer discounts to convert Monthly subscribers to Quarterly or Annual (e.g. "Switch to Annual and save 20%").

2. Target monthly subscribers who also show payment delays or low usage — that combination is the highest risk profile.

3. When acquiring new customers, incentivize longer contract commitment upfront.

Insight 4: New Customers (Low Tenure) Churn Most

Finding: Tenure is the 4th most important feature (score: 375). Customers with fewer months with the company have much higher churn rates.

Interpretation: The first 6 months are make-or-break. If a customer does not develop a habit or find sufficient value early, they leave. This is called **early-stage churn** and is common in subscription businesses.

Business action:

1. Invest heavily in **onboarding** — a structured 30/60/90 day program to help new customers get value quickly.
2. Send automated usage nudges to customers with low Usage_Frequency in their first 3 months.
3. Assign a customer success contact to new high-spend customers for their first quarter.

Insight 5: The Model Achieves 95.4% AUC With 99.56% Recall

Finding: The LightGBM model achieves AUC = 0.9541 and Recall = 0.9956 at threshold 0.15.

Interpretation:

- AUC of 0.95 means the model is near-optimal at ranking churners above non-churners.
- Recall of 99.56% means only 4 in 1,000 actual churners are missed.
- Precision of 89.65% means 9 in 10 customers flagged as "at risk" genuinely are.

Business action: This model is production-ready. At scale (505,207 customers), flagging all customers above 0.15 probability will:

- Identify the overwhelming majority of people about to churn
- Produce fewer than 11% false alarms
- Drive targeted, efficient retention campaigns

A typical churn model achieving 0.75 AUC would have far more missed churners. The 0.95+ result on this dataset means confidence in the predictions is high.

Insight 6: High-Risk Segment Revenue Is Concentrated

Finding: The `High Risk Revenue` measure (spend from customers with >70% churn probability) represents a disproportionate share of `Spend at Risk`.

Interpretation: High-risk customers are not uniformly distributed across spend levels. There is a concentration of high-spend customers in the high-risk category (consistent with Insight 1 above — spend is positively correlated with churn risk).

Business action:

Even a 10% retention conversion rate among high-risk customers recovers significant revenue (`Potential Spend Saved (20%)` shows the 20% scenario). A modest investment in a retention campaign — personal

outreach, discount offers, feature upgrades — pays for itself many times over.

What This Means For the Business: A Summary

| Problem | Signal in Data | Recommended Action |

|---|---|---|

| We are losing high-value customers | Total_Spend is top predictor | Proactively engage high-spenders showing any other warning signs |

| Payment issues lead to cancellation | Payment_Delay is 3rd predictor | Automate outreach at >15 day delay — offer flexibility |

| Monthly contracts have high churn | Contract_Length_Monthly is key | Incentivize upgrades to Quarterly/Annual |

| New customers leave before they bond | Tenure is 4th predictor | Invest in structured onboarding programs |

| Low usage = disengaged = leaving | Usage_Frequency is 5th predictor | Trigger re-engagement for customers using <5/month |

| Frustrated customers leave | Support_Calls is 6th predictor | Resolve issues in the first call; measure first-call resolution rate |

The ROI Case in Simple Terms

Assume:

- Average customer spend: ~$632
- High-risk customers: (read from Power BI dashboard)
- Retention campaign cost: ~$50 per contacted customer

For every 1,000 high-risk customers contacted:

- At 10% conversion: 100 customers saved × $632 = **$63,200 recovered**
- Campaign cost: 1,000 × $50 = $50,000
- **Net recovery: $13,200 at the conservative estimate**

At 20% conversion (still within realistic range based on literature):

- 200 customers saved × $632 = **$126,400 recovered**
- Campaign cost: $50,000
- **Net recovery: $76,400**

The model effectively turns a retention campaign from a wide-net guessing game into a laser-targeted, high-ROI operation.